

## E.1 / PAPER 2 / QUESTION 2

26 October 2023 / TZ2 (file label) / HL Paper 2 / original Q5 / 8823-6514

2. The table shows some of the energy levels for a hydrogen atom.

|                | Level        | Energy/eV |
|----------------|--------------|-----------|
| Ionized state  | $n = \infty$ | 0         |
| Excited states | $n = 3$      | -1.51     |
|                | $n = 2$      | -3.40     |
| Ground state   | $n = 1$      | -13.6     |

- (a) Distinguish between an atom in an excited state and an atom in an ionized state.

[1]

.....

.....

.....

.....

.....

.....

- (b) Explain how electromagnetic radiation is emitted from a hydrogen atom in an excited state.

[2]

.....

.....

.....

.....

.....

.....

(This question continues on the following page)

## E.1 / PAPER 2 / QUESTION 2

26 October 2023 / TZ2 (file label) / HL Paper 2 / original Q5 / 8823-6514

---

### (Question 2 continued)

- (c) When radiation of wavelength  $6.6 \times 10^{-7} \text{ m}$  is incident on cold hydrogen gas some of it is absorbed by the gas.

- (i) State the region of the electromagnetic spectrum of this radiation. [1]

.....

- (ii) Determine the initial and final states of the hydrogen atom that are involved in this absorption. [3]

.....  
.....  
.....  
.....  
.....  
.....